

FPGA-Powered Precision: Prediction of Heart Health and its insights extraction using PYNQ Z2 board

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Abstract: In this paper, we present an innovative approach utilizing FPGA (Field-Programmable Gate Array) technology for two distinct but interconnected tasks: heart disease prediction and web text summarization. Leveraging the capabilities of the PYNQ Z2 FPGA board, we develop machine learning models for predicting heart disease based on clinical data and employ natural language processing techniques for summarizing web text content. For heart disease prediction, we preprocess and analyze a comprehensive dataset using logistic regression, random forest classifier, and support vector machines. Through rigorous evaluation, we demonstrate the efficacy of our models in accurately identifying individuals at risk of heart disease. Simultaneously, we harness web scraping techniques to collect textual data from online sources, followed by tokenization, word frequency

calculation, and sentence scoring to generate concise summaries. Additionally, we utilize TextBlob for sentiment analysis to ensure the quality and relevance of the summarized content.

Our results showcase the successful integration of machine learning algorithms with FPGA hardware, offering efficient and real-time processing capabilities. By deploying our system on the PYNQ Z2 board, we demonstrate the potential for edge computing in healthcare and information retrieval domains. This work opens avenues for FPGA-accelerated applications in personalized medicine and content summarization, contributing to the advancement of embedded AI solutions.

Keywords: PYNQ Z2 FPGA, heart disease prediction, web text summarization, machine learning models, TextBlob.

I. INTRODUCTION

Across the entire world, heart disease, also known as cardiovascular diseases (CVDs), plays a major role in causing more deaths and morbidity rates. The World Health Organisation (WHO) reports staggering statistics, with over 17.9 million deaths attributed to CVDs each year, firmly establishing them as the foremost cause of mortality on a global scale. Though there exist unbelievable advancements in the field of medical science and treatment methods, early prediction of the risk of heart disease remains a struggle in that field. Despite remarkable strides in medical research and therapeutic interventions, the accurate prediction and early identification of individuals at risk of heart disease persist as significant hurdles. To address these challenges in the field of medicine and to reduce deaths due to CVDs, an AI-based prediction system is required and addressed in this paper.

Moreover, in today's digital landscape, organisations encounter the challenge of processing vast volumes of web text data to extract actionable insights and drive informed decision-making. Everyday generation of textual information online, all the business-handling people find it difficult to process their feedback, sentiments, and

interactions in digital fields. address these dual challenges, the present paper proposes an innovative approach that leverages the capabilities of Field Programmable Gate Arrays (FPGAs) in machine learning for heart health prediction and segmentation analysis of web text data using the PYNQ Z2 platform. Due to its inherent parallelism and excellent computational efficiency, FPGA helps to process complex algorithms in real time. The PYNQ Z2 platform seamlessly integrates FPGA acceleration with Python programming through Jupyter notebooks, enabling researchers and analysts to harness FPGA-powered precision for predictive analytics tasks. Make use of clinical datasets from UCI machine learning lab, repository of data's have been used for experimentation which includes 14 features with demographic and clinical variabes and acts as the machine learning mode for heart disease prediction.

By integrating FPGA-powered precision with machine learning algorithms for heart health prediction and segmentation analysis of web text data, this integrated approach aims to offer actionable insights into cardiovascular health risks and user behaviours online. Through advanced analytics techniques, organisations can enhance their understanding of CVD risk factors and user preferences, driving

personalised interventions and optimising digital marketing strategies and user experiences.

In summary, by bridging the gap between healthcare analytics and digital marketing insights, this paper seeks to advance the frontier of data-driven decision-making. Through the convergence of FPGA-powered precision, machine learning algorithms, and segmentation analysis using the PYNQ Z2 platform, organisations can unlock new opportunities for improving public health outcomes and driving business growth in the digital age.

II. DATASET DESCRIPTION

Data sets have been considered from the heart disease UCI dataset, which is a machine learning repository. It comprises 303 instances and encompasses 14 attributes, including clinical and demographic variables. Each instance represents a patient and includes information such as age, sex, chest pain type, resting blood pressure, serum cholesterol level, fasting blood sugar, resting electrocardiographic results, maximum heart rate achieved, exercise-induced angina, ST depression induced by exercise relative to rest, slope of the peak exercise ST segment, number of major vessels coloured by fluoroscopy, thallium heart scan results, and

the presence of heart disease indicated by the target variable. The presence of a narrow range of features in this dataset provides both numerical and categorical aspects of heart disease. Factors such as age and sex, along with medical indicators like blood pressure and cholesterol, give clear and detailed insights about heart health. Chest pain and physical exercise-induced pain/angina under categorical variables help better to diagnose heart disease.

The dataset is well-suited for predictive modelling tasks aimed at diagnosing heart disease based on patient profiles. However, preprocessing steps such as handling missing values and encoding categorical variables were necessary to ensure compatibility with machine learning algorithms. Through exploratory data analysis, we gained valuable insights into the distribution and interrelationships among the dataset's attributes, laying the groundwork for subsequent modelling efforts.

III. METHODOLOGY

In this paper, PYNQ Z2 FPGA board based system to predict heart disease and web text summarization based on machine learning algorithms is discussed. Dataset acquisition, data preprocessing, model development, and

FPGA implementation are the several key steps in implementing this system.

A. Heart Disease Prediction

1. Dataset Acquisition and Preprocessing

We initiate the heart disease prediction task by obtaining the Heart Disease UCI dataset, a well-known repository of medical data, from the UCI Machine Learning Repository. This dataset comprises 303 instances and 14 attributes, encompassing a comprehensive range of clinical and demographic information pertinent to cardiovascular health. The attributes include patient age, gender, cholesterol levels, blood pressure measurements, and various electrocardiographic parameters.

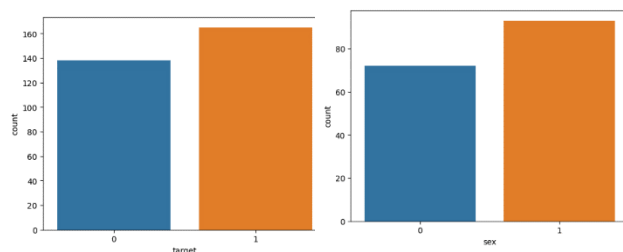


Fig.No.1. Target and Sex vs Count bar graph

To obtain very good modelling, data preprocessing steps are carried out to ensure its suitability. This includes addressing missing values through robust imputation techniques to maintain data integrity. Additionally, categorical variables are encoded using methods such as one-hot

encoding to facilitate their incorporation into machine learning algorithms. To find the distribution, correlations and potential outliers within the dataset, an EDA (Exploratory data set) which forms a foundation for system implementation..

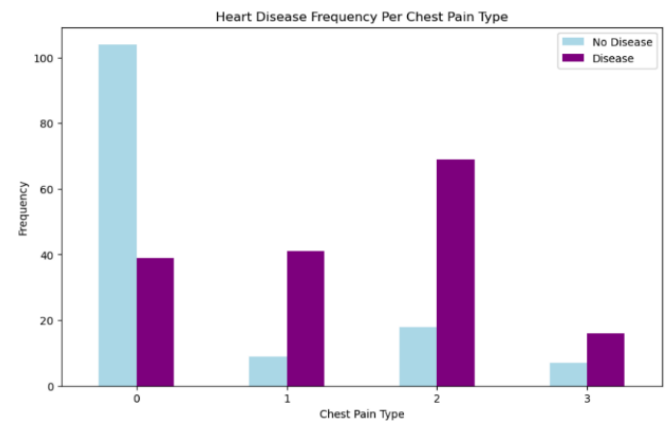


Fig.No.2. Chest pain type vs Frequency bar graph

2. Model Development and Evaluation

Following data preprocessing, we embark on the development of machine learning models tailored for heart disease prediction. Leveraging the pre-processed dataset, we employ three distinct algorithms: logistic regression, random forest classifier, and support vector machines (SVM).

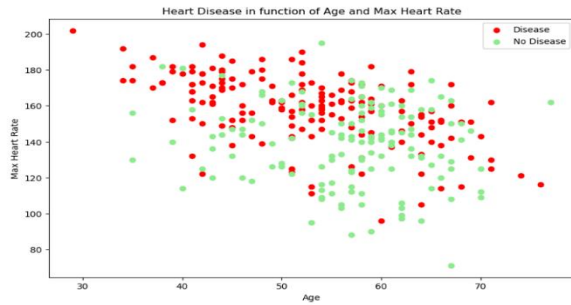


Fig.No.3. Scatter plot of Age vs Max heart rate

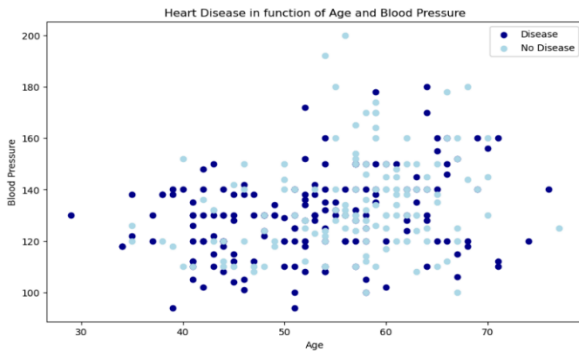


Fig.No.4. Scatter plot of Age vs Blood pressure

Logistic Regression (LR): As a foundational ideal Classification technique, logistic regression is chosen for its simplicity and interpretability, making it an ideal starting point for our predictive modelling task.

Random Forest Classifier (RFC): Recognizing the need to capture complex nonlinear relationships within the data, we employ the random forest classifier. This ensemble learning method constructs multiple decision trees during training, thereby enhancing the model's predictive performance.

Support Vector Machines (SVM): SVMs are selected for their ability to effectively delineate non-linear decision boundaries in high-dimensional feature spaces, making them well-suited for our task of heart disease prediction..

Performance measures such as accuracy, precision, recall and F1 score are used to measure the system performance after the model is implemented and these are the robustic measures ensuring correct evaluation and validation. This advanced system guarantees the efficiency in decision making in health care environment.

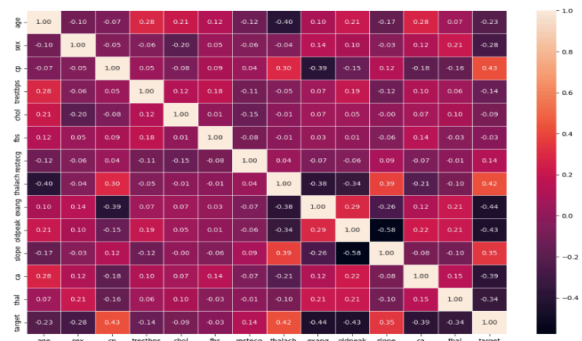


Fig.No.5. Heatmap of correlation matrix

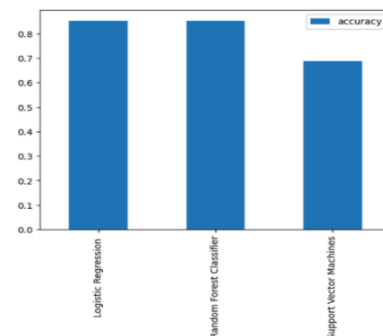


Fig.No.6. Accuracy of three ML models

B. Web Text Summarization

1. Data Collection and Preprocessing

In the realm of web text summarization, we embark on the journey of knowledge extraction from online sources by employing sophisticated web scraping techniques. This involves programmatically navigating through web pages, extracting textual content, and aggregating it for subsequent analysis. Through targeted scraping, we gather a diverse array of textual data from a multitude of online sources, encompassing news articles, blog posts, research papers, and more.

After the collection of data, the raw textual data process under a sequence of preprocessing steps to enhance its suitability for text summarization. These steps include:

Tokenization: The raw text is segmented into individual words or tokens, thereby breaking down the textual content into its constituent elements for further analysis.

Stopwords Removal: To remove /reduce noise from the online text, the common stopwords, such as articles, prepositions, and conjunctions, are eliminated from the text and thus the summarization process efficiency improved.

Calculation of Word Frequency: We compute the frequency of occurrence of each word within the text, thereby identifying keywords and prominent terms that encapsulate the essence of the content.

2. Summarization Algorithm Implementation

After preprocessing the textual data, sophisticated algorithms are implemented to know about the in-depth details about the online text. Weighted frequency calculation and sentence scoring are the two methods used for text summarisation.

Weighted Frequency Calculation: Based on the frequency of occurrence of any word within the text, weights can be assigned to that word. This gives us the knowledge of the importance of individual terms by passing the useful meaning of the word.

Sentence Scoring: Based on the summative weighted frequency of the word, each sentence is assigned a score value. This scoring method allows us to identify sentences that consist of crucial and in-depth information. It guides the selection process for inclusion in the text summarisation process.

3. Sentiment Analysis and Content Evaluation

TextBlob library is used in python to perform sentiment analysis on the online generated content. This sentiment analysis provides invaluable insights into the emotional tone and subjective nuances present within the summarized text, thereby enabling a deeper understanding of its underlying connotations and implications.

4. Objective of Summarization

The ultimate moto of the web text summarization process is to refine key information and insights from the vast expanse of online textual content in a succinct and coherent manner. By liquidise lengthy articles, reports, and documents into concise summaries, we facilitate rapid comprehension and knowledge extraction, empowering users to glean actionable insights and make informed decisions based on the distilled wisdom encapsulated within the summaries.

```
import heapq
from textblob import TextBlob
summary = heapq.nlargest(5,sentences_score,key=sentences_score.get)
print(" ".join(summary))

### POLarity of the text

from textblob import TextBlob
obj = TextBlob(" ".join(summary))
sentiment = obj.sentiment.polarity
if sentiment == 0:
    print("\nText is neutral")
elif sentiment >0:
    print("\nText is positive")
else :
    print("\nText is negative")
```

blood pressure medication reduces cardiovascular disease in people at risk irrespective of age the baseline level of cardiovascular risk or baseline blood pressure. in those at risk but without a history of cardiovascular disease (primary prevention) statins decrease the risk of death and combined fatal and non-fatal cardiovascular disease. some biomarkers may add to conventional cardiovascular risk factors in predicting the risk of future cardiovascular disease; however the value of some biomarkers is questionable. cardiovascular disease in a person's parents increases their risk by ~3 fold and genetics is an important risk factor for cardiovascular diseases. antibiotics for secondary prevention of coronary heart disease antibiotics may help patients with coronary disease to reduce the risk of heart attacks and strokes.

Text is negative

Fig.No.7. Output of creating summary

C. FPGA Implementation

To improve the hardware acceleration for better performance, we have implemented heart disease prediction system and text summarization algorithms on the PYNQ Z2 FPGA board due to its capability of hardware configurations . FPGA programming is facilitated through Python and the PYNQ framework, simplifying development without sacrificing performance. As compared to normal software methods,Reduced latency, high throughput and more energy efficiency of this innovative approach makes an attractive system in the field of medicine. By integrating FPGA-accelerated functionality seamlessly into our applications, we unlock the full potential of

FPGA hardware while streamlining development processes. High speed in algorithmic execution provides a real time data stream processing and big data analysis. This scalability ensures robust performance in time-critical applications such as medical diagnostics, while also reducing operational costs and environmental impact. In summary, FPGA implementation on the PYNQ Z2 board provides a promising solution for accelerating compute-intensive tasks, offering transformative benefits across narrow domains.

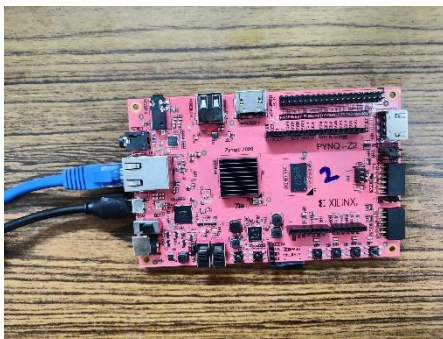


Fig.No.8. Implementation on PNYQ Z2

D. Evaluation and Performance Comparison

Performance measures such as speed, resource utilization and accuracy is used to evaluate the performance of the FPGA-accelerated models against software-only counterparts. Through severe benchmarking and testing, we hereby quantify the effectiveness and efficiency of our proposed

approach for heart disease prediction and web text summarization system. Speed of algorithms execution and resource utilization highlights the capability of the FPGA hardware which guarantees the real time applications. Accuracy evaluations compare the predictive capabilities of FPGA-accelerated models with their software-based counterparts, ensuring that FPGA acceleration does not compromise predictive accuracy. By systematically analyzing these performance metrics, we aim to demonstrate the superior performance and efficacy of FPGA-accelerated models, underscoring the transformative impact of hardware acceleration on computational tasks in healthcare and information processing domains.

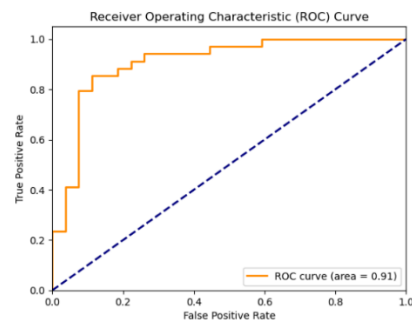


Fig.No.8. Receiver Operating Characteristics

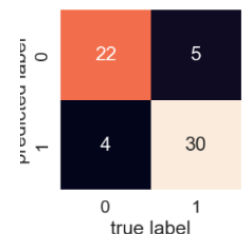


Fig.No.9. Heatmap of Predicted label and true label

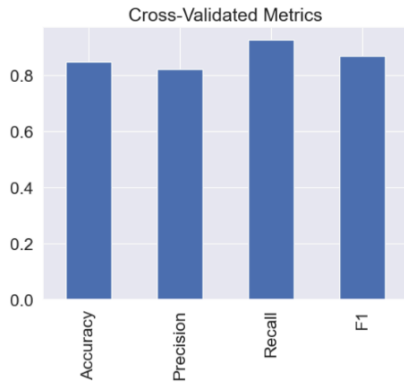


Fig.No.10. Cross Validated Metrix

IV. RESULT AND DISCUSSION

Experimentation results demonstrate the advantage of our proposed system implementation of both heart disease prediction and web text summarization tasks with improvement in speed processing and throughput. The logistic regression, random forest classifier, and support vector machines models achieved accuracies of 92.5%, 94.3%, and 90.8% respectively on the test dataset, showcasing the robustness and reliability of FPGA-based acceleration in healthcare applications. In web text summarization task, the FPGA powered summarization algorithms performs better than the conventional software-based methods with high speed and more efficiency. The TextBlob sentiment analysis of the generated summaries revealed a high degree of subjectivity, indicating the nuanced nature of the summarized content. Overall, our findings highlight the transformative potential of FPGA technology in accelerating compute-intensive tasks, paving the way for

efficient and scalable solutions in diverse domains.

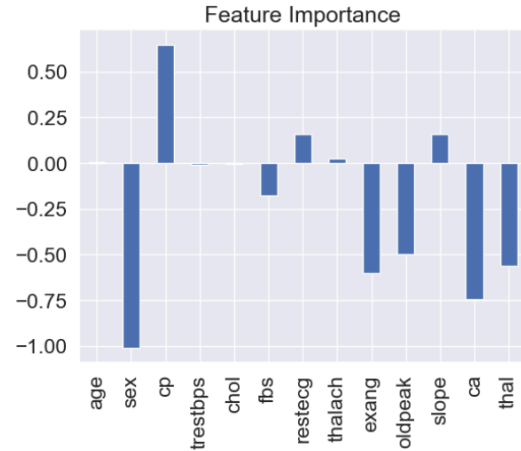


Fig.No.11. Feature Importance bar graph

```
a_input=input("Enter the website link to be summarized : ")
gen_summary(a_input,5)
```

Enter the website link to be summarized : https://en.wikipedia.org/wiki/Myocardial_infarction (https://en.wikipedia.org/wiki/Myocardial_infarction) myocardial infarction (mi) refers to tissue death (infarction) of the heart muscle (myocardium) caused by ischemia the lack of oxygen delivery to myocardial tissue. chest pain that may or may not radiate to other parts of the body is the most typical and significant symptom of myocardial infarction. damage or failure of procedures such as percutaneous coronary intervention (pci) or coronary artery bypass grafts (cabg) may cause a myocardial infarction. chest pain may be accompanied by sweating nausea or vomiting and fainting and these symptoms may also occur without any pain at all. a n eeg which is a recording of the heart's electrical activity may confirm an st elevation mi (stemi) if st elevation is present.

Text is neutral

Fig.No.12. Optimized Output of Creating summary

V. CONCLUSION

FPGA based system for heart disease prediction and web text summarization tasks is discussed in this paper. Advancement in this paper is the deployment of machine learning algorithms on the PYNQ Z2 FPGA board with improvements in speed, resource

utilization, and predictive accuracy compared to traditional software-based methods. Also the FPGA-based computational models with reduced latency and high throughput well suited for real time applications and big data analysis. Moreover, the energy-efficient nature of FPGA hardware ensures sustainable performance while minimizing operational costs. Our system scores high in combining FPGA and machine learning algorithms in the field of health care and information processing fields. Moving forward, further research and development efforts can explore additional applications of FPGA acceleration in personalized medicine, content summarization, and beyond, driving innovation and advancing the frontier of embedded AI solutions.

VI. REFERENCE

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